EngiQuoteKE

Master Build Blueprint

Version 4.0 • Agent-Ready • March 2026

BOQ Generation • Material Rate Engine • M-Pesa Payments • PDF Export

Project	EngiQuoteKE — SaaS Construction Cost Estimator (Kenya)
Domain	engiquote.co.ke / engiquote.com
Target Users	Contractors • Quantity Surveyors • Civil Engineers • Architects (Kenya)
Revenue Model	Free / Pro KSh 4,999/mo / Enterprise KSh 14,999/mo + PDF add-on KSh 999
Build Goal	First paying user in 6 weeks • 50 Pro subscribers in 60 days

1. Product Overview & Core Promise

EngiQuoteKE solves a real, expensive problem: Kenyan contractors, engineers and quantity surveyors waste billable hours producing manual estimates that are often inaccurate, outdated, and unprofessional. EngiQuoteKE replaces this with a browser-based tool that auto-generates a full Bill of Quantities (BOQ), applies live Kenyan market material rates including import duties, adds labour/profit/contingency margins automatically, and exports a branded client-ready PDF — in under 5 minutes.

1.1 What Makes This Different From the Previous Blueprints

The existing blueprints (v1–v3.5) gave a reasonable skeleton but left critical gaps that would cause an AI agent to stall or make wrong decisions. This blueprint fills those gaps:

- Exact file creation order with zero ambiguity
- Complete environment variable reference (every single one, with example values)
- Seeded material rates table — the agent can populate the database on Day 1
- Full M-Pesa Daraja 3.0 implementation including sandbox setup and callback handling
- Actual PDF template specification including layout, columns and branding rules
- Comprehensive error handling strategy for every integration point
- Deployment checklist including DNS, SSL and environment config on Vercel + Supabase
- Marketing launch playbook tied to the dev roadmap

1.2 Success Definition (Agent Knows When to Stop)

🚩 Definition of Done

A user can: (1) sign up with email or Google via Clerk, (2) enter a project — type, location, floor area, custom items, (3) click Generate Estimate and see a full BOQ with KSh totals in < 5 seconds, (4) download a branded PDF immediately, (5) upgrade to Pro via M-Pesa STK Push and have their subscription activate automatically, (6) return and see all saved projects on their dashboard. Everything else is secondary.

2. Technology Stack (Authoritative — No Substitutions)

Every library choice below is deliberate. An agent must NOT substitute alternatives without explicit human approval.

Layer	Technology	Why This One
Monorepo	Turborepo + pnpm	Shared packages (db, utils, ui) with fast caching. Zero npm/yarn.
Frontend	Next.js 15 App Router + TypeScript	React Server Components for fast calculator UI. Streaming support for PDF generation progress.
UI Library	Tailwind CSS + shadcn/ui + Radix	Pre-built accessible components (tables, modals, forms). shadcn is not a dependency — it's copied into /components.
Data Tables	TanStack Table v8	Client-side sorting, filtering and pagination for the BOQ table. Handles 200+ line items without performance issues.
API Layer	tRPC v11	End-to-end TypeScript types. No REST — no schema drift between client and server.
ORM + DB	Prisma 5 + PostgreSQL (Supabase)	ACID transactions for subscription state. Supabase gives Row Level Security out of the box.
Auth	Clerk (Organizations)	Handles team invites, roles, SSO, email verification. Webhook fires on user.created to insert our User row.
PDF Generation	@react-pdf/renderer (server-side)	Renders a React component to a PDF buffer on the server. Client never touches the PDF stream.
File Storage	Uploadthing	Stores generated PDF URLs. Returns a permanent public URL after upload.
Background Jobs	Inngest	Handles subscription renewal reminders, M-Pesa callback processing, email queuing.
Payments	M-Pesa Daraja 3.0 + Bank Transfer	STK Push is the primary flow. Bank transfer (manual verify) is the fallback for corporate users.
Search	Typesense (self-hosted, optional v2)	Full-text search over saved projects. Skip for MVP — add in Week 8.
Analytics	PostHog (self-hosted or cloud)	Funnel tracking: signup → calculator → generate → download → upgrade. Free tier sufficient for launch.
Error Tracking	Sentry	Capture calculator errors and payment failures immediately.
Rate Limiting	Upstash Redis	Serverless Redis. Limits free-tier users to 3 estimates/day.
Email	Resend + React Email	Transactional emails (welcome, subscription, renewal reminders). Free tier: 3,000 emails/month.
Hosting	Vercel (app) + Supabase (DB)	Vercel hobby is free to start. Upgrade to Pro (KSh 3,200/mo) when traffic grows. Supabase free tier covers MVP.

3. Monorepo Folder Structure (Complete)

Create this exact structure. Every file listed here must exist before the agent moves to the next section. Files marked [GENERATE] means the agent writes the code. Files marked [COPY] means copy from shadcn or official docs.

```
/engiquote-monorepo
├── apps/
│   └── engiquote/                  # Next.js 15 App
│       ├── app/
│       │   ├── (public)/
│       │   │   ├── page.tsx           [GENERATE] Landing page
│       │   │   ├── pricing/page.tsx   [GENERATE] Pricing tiers
│       │   │   └── about/page.tsx     [GENERATE] About page
│       │   ├── (auth)/
│       │   │   ├── sign-in/page.tsx   [COPY] Clerk SignIn component
│       │   │   └── sign-up/page.tsx   [COPY] Clerk SignUp component
│       │   ├── (app)/
│       │   │   ├── layout.tsx         [GENERATE] App shell with sidebar
│       │   │   ├── dashboard/page.tsx [GENERATE] Overview + recent projects
│       │   │   ├── calculator/page.tsx [GENERATE] Main BOQ tool
│       │   │   ├── projects/
│       │   │   │   ├── page.tsx       [GENERATE] Project list with search
│       │   │   │   └── [id]/page.tsx  [GENERATE] Single project view + PDF
│       │   │   ├── billing/page.tsx   [GENERATE] Subscription + payment history
│       │   │   └── settings/page.tsx  [GENERATE] Profile + company info
│       │   ├── (admin)/
│       │   │   ├── layout.tsx         [GENERATE] Admin guard (ADMIN role only)
│       │   │   └── rates/page.tsx     [GENERATE] Material rate editor
│       │   ├── api/
│       │   │   ├── trpc/[trpc]/route.ts [GENERATE] tRPC HTTP adapter
│       │   │   ├── webhooks/clerk/route.ts [GENERATE] Clerk user.created webhook
│       │   │   ├── webhooks/mpesa/route.ts [GENERATE] M-Pesa callback
│       │   │   └── inngest/route.ts   [GENERATE] Inngest handler
│       │   ├── layout.tsx             [GENERATE] Root layout + Clerk provider
│       │   └── globals.css
│       ├── components/
│       │   ├── ui/                    [COPY] shadcn components
│       │   ├── calculator/
│       │   │   ├── ProjectTypeSelector.tsx
│       │   │   ├── ItemsTable.tsx      [GENERATE] TanStack Table
│       │   │   ├── TotalsPanel.tsx
│       │   │   └── GenerateButton.tsx
│       │   ├── pdf/
│       │   │   └── BoqDocument.tsx    [GENERATE] react-pdf template
│       │   ├── dashboard/
```

```
│       │   │   ├── RecentProjects.tsx
│       │   │   └── StatsCards.tsx
│       │   └── shared/
│       │       ├── Navbar.tsx
│       │       ├── Sidebar.tsx
│       │       └── SubscriptionBadge.tsx
│       ├── lib/
│       │   ├── trpc-client.ts
│       │   ├── rate-limiter.ts
│       │   └── utils.ts
│       ├── server/
│       │   ├── trpc.ts              [GENERATE] tRPC init + context
│       │   └── routers/
│       │       ├── _app.ts          [GENERATE] Root router
│       │       ├── project.ts       [GENERATE]
│       │       ├── calculator.ts    [GENERATE]
│       │       ├── subscription.ts  [GENERATE]
│       │       ├── admin.ts         [GENERATE]
│       │       └── user.ts          [GENERATE]
│       └── next.config.ts
├── packages/
│   ├── db/
│   │   ├── prisma/
│   │   │   ├── schema.prisma        [GENERATE — Section 4]
│   │   │   └── seed.ts              [GENERATE — Section 5]
│   │   └── index.ts                 [GENERATE] Prisma client singleton
│   ├── utils/
│   │   ├── engi-calculator.ts       [GENERATE — Section 6]
│   │   ├── pdf-generator.ts         [GENERATE — Section 7]
│   │   ├── mpesa.ts                 [GENERATE — Section 9]
│   │   └── types.ts                 [GENERATE] Shared TypeScript types
│   ├── ui/                           [COPY] Shared design tokens
│   └── notifications/
│       ├── inngest.ts               [GENERATE] Inngest client
│       └── functions/
│           ├── renewal-reminder.ts  [GENERATE]
│           └── send-welcome.ts      [GENERATE]
├── turbo.json                        [GENERATE]
├── pnpm-workspace.yaml              [GENERATE]
└── .env.example                     [GENERATE — Section 10]
```

4. Complete Database Schema

File: packages/db/prisma/schema.prisma — Create this file FIRST before any other code. Run prisma migrate dev --name init immediately after.

```
generator client {
  provider = "prisma-client-js"
}

datasource db {
  provider = "postgresql"
  url      = env("DATABASE_URL")
  directUrl = env("DIRECT_URL")   // required for Supabase connection pooling
}

model User {
  id          String   @id @default(cuid())
  clerkId     String   @unique
  email       String   @unique
  companyName String?
  phone       String?  @unique
  role        Role     @default(ENGINEER)
  subscription Subscription?
  projects    Project[]
  createdAt   DateTime @default(now())
  updatedAt   DateTime @updatedAt
}

model Subscription {
  id          String    @id @default(cuid())
  userId      String    @unique
  plan        Plan      @default(FREE)
  status      SubStatus @default(ACTIVE)
  expiresAt   DateTime?
  mpesaRef    String?
  bankRef     String?
  projectsUsed Int      @default(0)  // track free tier usage
  createdAt   DateTime  @default(now())
  updatedAt   DateTime  @updatedAt
  user        User      @relation(fields: [userId], references: [id], onDelete:
Cascade)
}

model Project {
  id           String        @id @default(cuid())
```

```
  userId       String
  name         String
  type         ProjectType
  location     String?
  floorAreaSqm Decimal?
  totalCostKsh Decimal
  status       ProjectStatus @default(DRAFT)
  snapshot     Json          // full BOQ + rates snapshot for audit
  pdfUrl       String?
  sharedWith   String[]      // array of clerkIds
  includesDuties Boolean     @default(true)
  profitMargin Decimal       @default(0.20)
  notes        String?
  createdAt    DateTime      @default(now())
  updatedAt    DateTime      @updatedAt
  user         User          @relation(fields: [userId], references: [id],
onDelete: Cascade)
  boqItems     BoqItem[]
}


model BoqItem {
  id          String  @id @default(cuid())
  projectId   String
  section     String  @default("General")  // e.g. Substructure, Superstructure
  description String
  quantity    Decimal
  unit        String
  unitPrice   Decimal
  totalPrice  Decimal
  isCustom    Boolean @default(false)
  sortOrder   Int     @default(0)
  project     Project @relation(fields: [projectId], references: [id], onDelete:
Cascade)
}


model MaterialRate {
  id             String   @id @default(cuid())
  category       String   @unique // e.g. cement-bag, steel-rebar-ton
  displayName    String   // e.g. "Cement (50kg bag)"
  section        String   // e.g. "Walling", "Roofing", "Substructure"
  unit           String   // e.g. bag, ton, sqm, kg, m3
  rateKsh        Decimal
  importDutyRate Decimal? // e.g. 0.25 = 25% duty
  isImported     Boolean  @default(false)
  isActive       Boolean  @default(true)
  notes          String?
  lastUpdated    DateTime @updatedAt
```

```
}

model PaymentLog {
  id           String   @id @default(cuid())
  userId       String
  amount       Decimal
  currency     String   @default("KES")
  method       String   // mpesa | bank
  reference    String?  // mpesa transaction ID or bank ref
  status       String   // pending | success | failed
  plan         Plan
  rawResponse  Json?    // full Daraja API response
  createdAt    DateTime @default(now())
}

enum Role { ENGINEER QUANTITY_SURVEYOR CONTRACTOR ARCHITECT ADMIN }
enum Plan { FREE PRO ENTERPRISE }
enum SubStatus { ACTIVE PAST_DUE CANCELLED TRIALING }
enum ProjectType { BUILDING ROAD BRIDGE RENOVATION INTERIOR_FITOUT CIVIL }
enum ProjectStatus { DRAFT COMPLETED SHARED ARCHIVED }
```

⚑ Why PaymentLog is New

The previous blueprints had no payment audit trail. PaymentLog records every M-Pesa attempt — including failures — so you can debug payment issues, issue refunds, and reconcile against Safaricom statements.

5. Database Seed — Material Rates (Day 1 Data)

File: packages/db/prisma/seed.ts — Run this immediately after the first migration so the calculator has live data. These rates are current as of March 2026 (Nairobi market). The admin panel allows updating without redeployment.

```
// packages/db/prisma/seed.ts
import { PrismaClient } from '@prisma/client';
const prisma = new PrismaClient();


const RATES = [
  // ── SUBSTRUCTURE ──────────────────────────────────────────────
  { category: 'excavation-m3',       displayName: 'Excavation (per m³)',
section: 'Substructure', unit: 'm3',  rateKsh: 800,    isImported: false },
  { category: 'hardcore-m3',          displayName: 'Hardcore filling (m³)',
section: 'Substructure', unit: 'm3',  rateKsh: 2200,   isImported: false },
  { category: 'blinding-concrete-m3', displayName: 'Blinding concrete (m³)',
section: 'Substructure', unit: 'm3',  rateKsh: 14000,  isImported: false },
  { category: 'strip-foundation-m3', displayName: 'Strip foundation concrete',
section: 'Substructure', unit: 'm3',  rateKsh: 16500,  isImported: false },


  // ── WALLING ───────────────────────────────────────────────────
  { category: 'cement-bag',           displayName: 'Cement (50kg bag)',
section: 'Walling',      unit: 'bag', rateKsh: 900,    isImported: false },
  { category: 'sand-m3',              displayName: 'River sand (m³)',
section: 'Walling',      unit: 'm3',  rateKsh: 2800,   isImported: false },
  { category: 'ballast-m3',           displayName: 'Ballast / Aggregate (m³)',
section: 'Walling',      unit: 'm3',  rateKsh: 3200,   isImported: false },
  { category: 'burnt-brick-each',     displayName: 'Burnt brick (each)',
section: 'Walling',      unit: 'pce', rateKsh: 18,     isImported: false },
  { category: 'hollow-block-6inch',   displayName: 'Hollow block 6" (each)',
section: 'Walling',      unit: 'pce', rateKsh: 75,     isImported: false },


  // ── STEEL / STRUCTURAL ────────────────────────────────────────
  { category: 'steel-rebar-y12',      displayName: 'Y12 Steel rebar (ton)',
section: 'Structural',   unit: 'ton', rateKsh: 105000, isImported: true,
importDutyRate: 0.25 },
  { category: 'steel-rebar-y16',      displayName: 'Y16 Steel rebar (ton)',
section: 'Structural',   unit: 'ton', rateKsh: 102000, isImported: true,
importDutyRate: 0.25 },
  { category: 'brc-mesh-a142',        displayName: 'BRC Mesh A142 (sqm)',
section: 'Structural',   unit: 'sqm', rateKsh: 750,    isImported: true,
importDutyRate: 0.20 },
  { category: 'binding-wire-kg',      displayName: 'Binding wire (kg)',
section: 'Structural',   unit: 'kg',  rateKsh: 230,    isImported: false },


  // ── ROOFING ───────────────────────────────────────────────────
  { category: 'roofing-sheet-gauge30', displayName: 'Gauge 30 iron sheet (sqm)',
section: 'Roofing',    unit: 'sqm', rateKsh: 850,    isImported: true,
importDutyRate: 0.15 },
```

```
  { category: 'roofing-sheet-gauge28', displayName: 'Gauge 28 iron sheet (sqm)',
section: 'Roofing',    unit: 'sqm', rateKsh: 1100,   isImported: true,
importDutyRate: 0.15 },
  { category: 'roofing-timber-4x2',   displayName: 'Timber 4x2 (per metre)',
section: 'Roofing',      unit: 'm',   rateKsh: 180,    isImported: false },
  { category: 'ridge-cap-lm',         displayName: 'Ridge cap (lm)',
section: 'Roofing',      unit: 'm',   rateKsh: 420,    isImported: false },


  // ── FINISHES ──────────────────────────────────────────────
  { category: 'floor-tiles-sqm',      displayName: 'Floor tiles (sqm)',
section: 'Finishes',     unit: 'sqm', rateKsh: 1800,   isImported: true,
importDutyRate: 0.25 },
  { category: 'wall-plaster-sqm',     displayName: 'Wall plaster (sqm)',
section: 'Finishes',     unit: 'sqm', rateKsh: 380,    isImported: false },
  { category: 'paint-exterior-litre', displayName: 'Exterior paint (litre)',
section: 'Finishes',     unit: 'l',   rateKsh: 650,    isImported: false },
  { category: 'paint-interior-litre', displayName: 'Interior paint (litre)',
section: 'Finishes',     unit: 'l',   rateKsh: 480,    isImported: false },


  // ── WINDOWS & DOORS ───────────────────────────────────────
  { category: 'aluminium-window-sqm', displayName: 'Aluminium window (sqm)',
section: 'Openings',     unit: 'sqm', rateKsh: 7500,   isImported: true,
importDutyRate: 0.20 },
  { category: 'steel-door-each',      displayName: 'Steel door (each)',
section: 'Openings',     unit: 'pce', rateKsh: 18000,  isImported: true,
importDutyRate: 0.25 },
  { category: 'flush-door-each',      displayName: 'Flush door (each)',
section: 'Openings',     unit: 'pce', rateKsh: 8500,   isImported: false },


  // ── PLUMBING ──────────────────────────────────────────────
  { category: 'pvc-pipe-110mm-lm',   displayName: 'PVC pipe 110mm (lm)',
section: 'Plumbing',     unit: 'm',   rateKsh: 620,    isImported: false },
  { category: 'wc-suite-each',        displayName: 'WC suite (each)',
section: 'Plumbing',     unit: 'pce', rateKsh: 12000,  isImported: true,
importDutyRate: 0.25 },
  { category: 'basin-each',           displayName: 'Wash basin (each)',
section: 'Plumbing',     unit: 'pce', rateKsh: 6500,   isImported: true,
importDutyRate: 0.25 },
];


async function main() {
  console.log('Seeding material rates...');
  for (const rate of RATES) {
    await prisma.materialRate.upsert({
      where: { category: rate.category },
      update: rate,
      create: rate,
    });
  }
  console.log(`Seeded ${RATES.length} rates.`);
}
```

```
main().catch(console.error).finally(() => prisma.$disconnect());
```

6. Calculator Engine (Authoritative Implementation)

File: packages/utils/engi-calculator.ts — This is the core money-making function. Implement it exactly as written. Every line is intentional.

6.1 Formula Reference

Component	Formula
Unit Price (with duty)	rate.rateKsh × (1 + importDutyRate)
Item Total	unitPrice × quantity
Subtotal	Σ all item totals
Labour	subtotal × 0.25 (25% — editable)
Contingency	subtotal × 0.08 (8% — editable)
Profit Base	subtotal + labour + contingency
Profit	profitBase × profitMargin (default 20%)
Grand Total	subtotal + labour + contingency + profit

6.2 Implementation

```
// packages/utils/engi-calculator.ts
import { prisma } from '@engiquote/db';
import { Decimal } from '@prisma/client/runtime/library';
import { z } from 'zod';


export const CalculatorInputSchema = z.object({
  projectName:     z.string().min(2).max(120),
  projectType:
z.enum(['BUILDING','ROAD','BRIDGE','RENOVATION','INTERIOR_FITOUT','CIVIL']),
  location:        z.string().optional(),
  floorAreaSqm:    z.number().positive().optional(),
  includeImportDuties: z.boolean().default(true),
  labourRate:      z.number().min(0.10).max(0.50).default(0.25),
  contingencyRate: z.number().min(0).max(0.20).default(0.08),
  profitMargin:    z.number().min(0).max(0.40).default(0.20),
  customItems: z.array(z.object({
    category:    z.string(),          // must match MaterialRate.category
    description: z.string(),
    quantity:    z.number().positive(),
    unit:        z.string(),
    section:     z.string().default('Custom'),
  })).default([]),
```

```
  useTemplate: z.boolean().default(true),  // auto-add standard items from type
});

export type CalculatorInput = z.infer<typeof CalculatorInputSchema>;

export async function calculateEstimate(input: CalculatorInput) {
  const data = CalculatorInputSchema.parse(input);

  // 1. Fetch all active material rates once
  const rates = await prisma.materialRate.findMany({ where: { isActive: true } });
  const rateMap = new Map(rates.map(r => [r.category, r]));

  let subtotal = new Decimal(0);
  const boqItems: BoqLineItem[] = [];

  // 2. Apply project template items
  if (data.useTemplate) {
    const templateItems = getTemplateItems(data.projectType, data.floorAreaSqm);
    for (const item of templateItems) {
      const rate = rateMap.get(item.category);
      if (!rate) continue;
      const result = applyRate(rate, item.quantity, data.includeImportDuties);
      subtotal = subtotal.add(result.totalPrice);
      boqItems.push({ ...item, ...result, isCustom: false });
    }
  }

  // 3. Apply custom items
  for (const custom of data.customItems) {
    const rate = rateMap.get(custom.category);
    if (!rate) {
      // Custom item with no DB rate — skip (front-end should validate this)
      continue;
    }
    const result = applyRate(rate, custom.quantity, data.includeImportDuties);
    subtotal = subtotal.add(result.totalPrice);
    boqItems.push({
      category: custom.category,
      description: custom.description,
      quantity: new Decimal(custom.quantity),
      unit: custom.unit,
      section: custom.section,
      ...result,
      isCustom: true,
    });
  }
```

```
  // 4. Apply labour, contingency, profit
  const labour      = subtotal.mul(data.labourRate);
  const contingency = subtotal.mul(data.contingencyRate);
  const profitBase  = subtotal.add(labour).add(contingency);
  const profit      = profitBase.mul(data.profitMargin);
  const grandTotal  = profitBase.add(profit);

  return {
    boqItems,
    subtotal,
    labour,
    contingency,
    profit,
    grandTotal,
    meta: {
      labourRate: data.labourRate,
      contingencyRate: data.contingencyRate,
      profitMargin: data.profitMargin,
      includesImportDuties: data.includeImportDuties,
      generatedAt: new Date().toISOString(),
    }
  };
}

function applyRate(rate: any, quantity: number, includeDuties: boolean) {
  let unitPrice = new Decimal(rate.rateKsh);
  if (includeDuties && rate.importDutyRate && rate.isImported) {
    unitPrice = unitPrice.mul(new Decimal(1).add(rate.importDutyRate));
  }
  const totalPrice = unitPrice.mul(new Decimal(quantity));
  return { unitPrice, totalPrice };
}
```

6.3 Project Templates (Minimum Viable)

File: packages/utils/templates.ts — These auto-populate the BOQ based on project type. Agent must implement all six types.

```
// Returns standard items for a given project type
// floorAreaSqm is used to scale quantities for BUILDING type
export function getTemplateItems(type: string, floorAreaSqm?: number) {
  const area = floorAreaSqm ?? 100; // default 100sqm if not provided

  const templates: Record<string, TemplateItem[]> = {
    BUILDING: [
```

```
      { category: 'excavation-m3',        description: 'Excavation for strip
foundations', quantity: area * 0.3,  unit: 'm3',  section: 'Substructure' },
      { category: 'cement-bag',            description: 'Cement (walling)',
quantity: area * 0.8,  unit: 'bag', section: 'Walling' },
      { category: 'steel-rebar-y12',       description: 'Y12 Steel rebar',
quantity: area * 0.02, unit: 'ton', section: 'Structural' },
      { category: 'roofing-sheet-gauge30', description: 'Gauge 30 roofing sheets',
quantity: area * 1.15, unit: 'sqm', section: 'Roofing' },
      { category: 'floor-tiles-sqm',       description: 'Floor tiles',
quantity: area,        unit: 'sqm', section: 'Finishes' },
      { category: 'wall-plaster-sqm',      description: 'Internal wall plaster',
quantity: area * 3.5,  unit: 'sqm', section: 'Finishes' },
    ],
    RENOVATION: [
      { category: 'wall-plaster-sqm',      description: 'Replaster walls',
quantity: area * 2,   unit: 'sqm', section: 'Finishes' },
      { category: 'floor-tiles-sqm',       description: 'Replace floor tiles',
quantity: area,       unit: 'sqm', section: 'Finishes' },
      { category: 'paint-interior-litre',  description: 'Interior repaint',
quantity: area * 0.3, unit: 'l',   section: 'Finishes' },
    ],
    // ROAD, BRIDGE, INTERIOR_FITOUT, CIVIL — agent fills these in same pattern
  };


  return templates[type] ?? [];
}
```

7. PDF Generation (Exact Template Specification)

The previous blueprints said 'generate a PDF' without specifying what goes in it. This section describes the exact layout so the output looks professional enough to hand to a client.

7.1 PDF Layout Specification

Page size	A4 portrait (595 × 842 pt)
Margins	40pt top/bottom, 30pt left/right
Primary font	Helvetica (built into react-pdf, no import needed)
Brand colour	#1A3C5E (dark navy)
Accent colour	#E8622A (orange)
Header height	80pt — company logo (if provided) + EngiQuoteKE badge
Footer	Page X of Y • Generated on [date] • engiquote.co.ke
BOQ table columns	Section / Description / Qty / Unit / Unit Price KSh / Total KSh
Column widths	60 / 180 / 40 / 35 / 85 / 85 (total 485pt)
Section rows	Full-width dark blue row with white section name
Totals section	Right-aligned: Subtotal / Labour / Contingency / Profit / GRAND TOTAL
Grand total row	Bold, orange background, large font (14pt)
Disclaimer text	8pt gray: 'Rates valid as of [date]. Subject to market variation.'

7.2 Implementation

```
// packages/utils/pdf-generator.ts
import React from 'react';
import { Document, Page, Text, View, StyleSheet, pdf } from '@react-pdf/renderer';
import { utapi } from './uploadthing-server'; // server-side Uploadthing client

// Agent: implement the full PDF using these styles as a guide
const styles = StyleSheet.create({
  page:           { fontFamily: 'Helvetica', fontSize: 9, padding: '40 30' },
  header:         { flexDirection: 'row', justifyContent: 'space-between',
marginBottom: 20 },
  headerTitle:    { fontSize: 18, fontFamily: 'Helvetica-Bold', color: '#1A3C5E'
},
  headerSub:      { fontSize: 9, color: '#718096', marginTop: 2 },
  sectionRow:     { backgroundColor: '#1A3C5E', padding: '4 6', marginTop: 8 },
  sectionText:    { color: 'white', fontFamily: 'Helvetica-Bold', fontSize: 9 },
```

```
  tableHeader:    { flexDirection: 'row', backgroundColor: '#E8F0F7', padding: '4
0' },
  tableRow:       { flexDirection: 'row', borderBottomWidth: 0.5,
borderBottomColor: '#E2E8F0', padding: '3 0' },
  col_section:    { width: 60 },
  col_desc:       { width: 180 },
  col_qty:        { width: 40, textAlign: 'right' },
  col_unit:       { width: 35, textAlign: 'center' },
  col_unitprice:  { width: 85, textAlign: 'right' },
  col_total:      { width: 85, textAlign: 'right' },
  totalsContainer:{ marginTop: 16, alignItems: 'flex-end' },
  totalRow:       { flexDirection: 'row', justifyContent: 'flex-end', marginTop: 2
},
  totalLabel:     { width: 150, textAlign: 'right', paddingRight: 8, color:
'#4A5568' },
  totalValue:     { width: 90, textAlign: 'right', fontFamily: 'Helvetica-Bold' },
  grandTotalRow:  { backgroundColor: '#E8622A', padding: '5 6', marginTop: 4,
flexDirection: 'row', justifyContent: 'flex-end' },
  grandTotalLabel:{ width: 150, textAlign: 'right', paddingRight: 8, color:
'white', fontFamily: 'Helvetica-Bold', fontSize: 11 },
  grandTotalValue:{ width: 90, textAlign: 'right', color: 'white', fontFamily:
'Helvetica-Bold', fontSize: 11 },
  disclaimer:     { marginTop: 24, fontSize: 7, color: '#A0AEC0' },
  footer:         { position: 'absolute', bottom: 20, left: 30, right: 30,
flexDirection: 'row', justifyContent: 'space-between' },
});


export async function generatePdf(data: PdfData): Promise<string> {
  const doc = React.createElement(BoqDocument, { data });
  const pdfBuffer = await pdf(doc).toBuffer();


  // Upload to Uploadthing
  const file = new File([pdfBuffer], `${data.projectName.replace(/\s+/g,'-
')}.pdf`, {
    type: 'application/pdf'
  });
  const response = await utapi.uploadFiles([file]);
  if (!response[0]?.data?.url) throw new Error('PDF upload failed');
  return response[0].data.url;
}
```

8. API — Complete tRPC Router Specification

All procedures are listed here. The agent must implement every one. Procedures marked [PROTECTED] require Clerk auth middleware. Procedures marked [ADMIN] require role === ADMIN.

Procedure	Type	Description
calculator.generate	mutation [PROTECTED]	Runs calculateEstimate(), saves result as Draft project, returns full BOQ + PDF URL
project.getAll	query [PROTECTED]	Returns all projects for current user. Accepts optional status filter and search term.
project.getById	query [PROTECTED]	Returns single project including boqItems. Checks userId or sharedWith.
project.updateStatus	mutation [PROTECTED]	Updates ProjectStatus (DRAFT → COMPLETED → ARCHIVED).
project.delete	mutation [PROTECTED]	Soft deletes project (sets status ARCHIVED). Decrements subscription.projectsUsed.
project.share	mutation [PROTECTED]	Adds clerkIds to project.sharedWith array (Pro only).
project.regeneratePdf	mutation [PROTECTED]	Re-runs PDF generation on existing project (uses stored snapshot).
subscription.getCurrent	query [PROTECTED]	Returns current user's Subscription including plan, status, expiresAt.
subscription.upgrade	mutation [PROTECTED]	Creates pending Subscription + PaymentLog, initiates M-Pesa STK Push or returns bank details.
subscription.checkFreeTierLimit	query [PROTECTED]	Returns { allowed: boolean, used: number, limit: 3 } for free tier gate.
admin.getRates	query [ADMIN]	Returns all MaterialRates including inactive ones.
admin.updateRate	mutation [ADMIN]	Updates a MaterialRate by id. Validates all fields with Zod.
admin.createRate	mutation [ADMIN]	Creates a new MaterialRate. category must be unique.
admin.toggleRate	mutation [ADMIN]	Toggles isActive on a MaterialRate.
admin.getPaymentLogs	query [ADMIN]	Returns PaymentLogs with pagination. Used for revenue dashboard.
admin.manualActivate	mutation [ADMIN]	Manually activates a subscription (used for bank transfer verification).
user.getProfile	query [PROTECTED]	Returns current User with company info.

Procedure	Type	Description
user.updateProfile	mutation [PROTECTED]	Updates companyName, phone. Does not touch clerkId.

9. M-Pesa Daraja 3.0 — Full Implementation

This is the section that previous blueprints glossed over. Every step is specified here because payment failures are the #1 reason SaaS projects in Kenya fail to monetise.

9.1 Setup (Safaricom Developer Portal)

1. Go to developer.safaricom.co.ke and create an app
2. In sandbox mode, use Test credentials: shortcode 174379, passkey from portal
3. Set callback URL to https://engiquote.co.ke/api/webhooks/mpesa (must be HTTPS)
4. For production: apply for Go Live after testing at least 3 successful sandbox transactions
5. Use Ngrok locally for callback testing: ngrok http 3000 → copy HTTPS URL → set as callback

9.2 STK Push Flow

```
// packages/utils/mpesa.ts
import axios from 'axios';

const MPESA_BASE = process.env.NODE_ENV === 'production'
  ? 'https://api.safaricom.co.ke'
  : 'https://sandbox.safaricom.co.ke';

async function getAccessToken(): Promise<string> {
  const auth = Buffer.from(
    `${process.env.MPESA_CONSUMER_KEY}:${process.env.MPESA_CONSUMER_SECRET}`
  ).toString('base64');

  const { data } = await axios.get(
    `${MPESA_BASE}/oauth/v1/generate?grant_type=client_credentials`,
    { headers: { Authorization: `Basic ${auth}` } }
  );
  return data.access_token;
}

export async function initiateSTKPush({
  phone, amount, orderId, userId
}: { phone: string; amount: number; orderId: string; userId: string }) {
  const token = await getAccessToken();
  const timestamp = new Date().toISOString().replace(/[^0-9]/g, '').slice(0, 14);
  const password = Buffer.from(
    `${process.env.MPESA_SHORTCODE}${process.env.MPESA_PASSKEY}${timestamp}`
  ).toString('base64');
```

```
  const payload = {
    BusinessShortCode: process.env.MPESA_SHORTCODE,
    Password: password,
    Timestamp: timestamp,
    TransactionType: 'CustomerPayBillOnline',
    Amount: Math.ceil(amount),          // M-Pesa only accepts integers
    PartyA: phone.replace(/^0/, '254'), // normalise: 0712... → 254712...
    PartyB: process.env.MPESA_SHORTCODE,
    PhoneNumber: phone.replace(/^0/, '254'),
    CallBackURL: `${process.env.NEXT_PUBLIC_APP_URL}/api/webhooks/mpesa`,
    AccountReference: `EngiQuote-${orderId}`,
    TransactionDesc: `EngiQuoteKE Pro Subscription`,
  };

  const { data } = await axios.post(
    `${MPESA_BASE}/mpesa/stkpush/v1/processrequest`,
    payload,
    { headers: { Authorization: `Bearer ${token}` } }
  );

  return data; // { MerchantRequestID, CheckoutRequestID, ResponseCode, ... }
}
```

9.3 Callback Handler

```
// app/api/webhooks/mpesa/route.ts
import { NextRequest, NextResponse } from 'next/server';
import { prisma } from '@engiquote/db';

export async function POST(req: NextRequest) {
  const body = await req.json();
  const callback = body?.Body?.stkCallback;

  if (!callback) return NextResponse.json({ ok: false }, { status: 400 });

  const { ResultCode, MerchantRequestID, CallbackMetadata } = callback;

  // Find the pending PaymentLog by MerchantRequestID
  const log = await prisma.paymentLog.findFirst({
    where: { reference: MerchantRequestID, status: 'pending' }
  });

  if (!log) return NextResponse.json({ ok: true }); // ignore duplicates

  if (ResultCode === 0) {
```

```
    // Payment succeeded
    const mpesaRef = CallbackMetadata?.Item?.find((i: any) => i.Name ===
'MpesaReceiptNumber')?.Value;

    await prisma.$transaction([
      prisma.paymentLog.update({
        where: { id: log.id },
        data: { status: 'success', reference: mpesaRef, rawResponse: body }
      }),
      prisma.subscription.update({
        where: { userId: log.userId },
        data: {
          plan: log.plan,
          status: 'ACTIVE',
          mpesaRef,
          expiresAt: new Date(Date.now() + 30 * 24 * 60 * 60 * 1000), // +30 days
        }
      })
    ]);
  } else {
    // Payment failed (user cancelled, insufficient funds, timeout)
    await prisma.paymentLog.update({
      where: { id: log.id },
      data: { status: 'failed', rawResponse: body }
    });
  }

  return NextResponse.json({ ok: true });
}
```

⚑ Critical: Never trust M-Pesa ResultCode alone

Always verify the amount in CallbackMetadata matches the expected plan price before activating the subscription. A user paying KSh 1 for a Pro plan is a real attack vector. Verify: Item[Amount].Value === 4999 (or 14999 for Enterprise).

10. Complete Environment Variables Reference

File: .env.example — Copy this to .env.local for development. NEVER commit .env.local to git. Every variable is required unless marked [optional].

```
# ── DATABASE ──────────────────────────────────────────────────
DATABASE_URL="postgresql://postgres:[password]@db.[ref].supabase.co:5432/postgres?pgbounce
r=true"
DIRECT_URL="postgresql://postgres:[password]@db.[ref].supabase.co:5432/postgres"
# Get both from Supabase Dashboard → Settings → Database → Connection string

# ── AUTHENTICATION (Clerk) ────────────────────────────────────
NEXT_PUBLIC_CLERK_PUBLISHABLE_KEY="pk_live_..."
CLERK_SECRET_KEY="sk_live_..."
CLERK_WEBHOOK_SECRET="whsec_..."
# Webhook secret: Clerk Dashboard → Webhooks → Signing Secret
# Required events: user.created, user.deleted

# ── M-PESA DARAJA ─────────────────────────────────────────────
MPESA_CONSUMER_KEY="..."
MPESA_CONSUMER_SECRET="..."
MPESA_SHORTCODE="174379"            # Sandbox: 174379 | Production: your paybill
MPESA_PASSKEY="bfb279f9aa9bdbcf1..." # From Safaricom developer portal
MPESA_ENV="sandbox"                  # Change to 'production' for go-live

# ── FILE STORAGE (Uploadthing) ────────────────────────────────
UPLOADTHING_TOKEN="eyJ..."
# Get from uploadthing.com → Your App → API Keys

# ── BACKGROUND JOBS (Inngest) ─────────────────────────────────
INNGEST_SIGNING_KEY="signkey-prod-..."
INNGEST_EVENT_KEY="..."

# ── RATE LIMITING (Upstash Redis) ─────────────────────────────
UPSTASH_REDIS_REST_URL="https://...upstash.io"
UPSTASH_REDIS_REST_TOKEN="..."

# ── EMAIL (Resend) ────────────────────────────────────────────
RESEND_API_KEY="re_..."
EMAIL_FROM="noreply@engiquote.co.ke"

# ── ANALYTICS (PostHog) [optional for MVP] ────────────────────
NEXT_PUBLIC_POSTHOG_KEY="phc_..."
NEXT_PUBLIC_POSTHOG_HOST="https://app.posthog.com"
```

```
# ── ERROR TRACKING (Sentry) [optional for MVP] ────────────────────
SENTRY_DSN="https://...@sentry.io/..."
NEXT_PUBLIC_SENTRY_DSN="https://...@sentry.io/..."

# ── APP CONFIG ────────────────────────────────────────────────────
NEXT_PUBLIC_APP_URL="https://engiquote.co.ke"  # http://localhost:3000 for dev
ADMIN_CLERK_ID="user_..."   # Your Clerk user ID — gives ADMIN role in DB
```

11. Agent Build Order (No Deviation Permitted)

This is the most important section. Every previous blueprint gave a rough week-by-week plan. This section gives an exact ordered task list — the agent must complete each task and verify it works before moving to the next. Do NOT parallelise these steps.

> **🚩 Agent Rule**
> After completing each numbered task, run the verification command listed. Only move to the next task when verification passes. If a task cannot be verified, stop and report the blocker.

Phase 1: Foundation (Days 1–3)

Task 1.1	Create GitHub repo engiquote-monorepo. Init Turborepo with pnpm. Create folder structure exactly as in Section 3.
Verify	pnpm install completes with zero errors. turbo build passes on empty workspace.
Task 1.2	Copy Section 4 schema into packages/db/prisma/schema.prisma. Configure Supabase project. Run: pnpm prisma migrate dev --name init
Verify	Supabase dashboard shows all 6 tables (User, Subscription, Project, BoqItem, MaterialRate, PaymentLog).
Task 1.3	Copy seed.ts from Section 5. Run: pnpm prisma db seed
Verify	SELECT COUNT(*) FROM "MaterialRate"; returns 28+ rows in Supabase SQL editor.
Task 1.4	Install and configure Clerk. Create sign-in and sign-up pages. Create the Clerk webhook handler (/api/webhooks/clerk) that creates a User row on user.created.
Verify	Sign up with a test email. Check Supabase — a User row appears automatically.

Phase 2: Calculator (Days 4–7)

Task 2.1	Implement engi-calculator.ts from Section 6. Implement templates.ts with at least BUILDING and RENOVATION templates.
Verify	Write a test: calculateEstimate({ projectName: 'Test', projectType: 'BUILDING', floorAreaSqm: 100, ... }). Grand total should be > KSh 1,000,000 for 100sqm.
Task 2.2	Set up tRPC. Implement calculator.generate procedure. Connect to the calculator engine.
Verify	Call the procedure directly from a test script. Receive a valid result object with grandTotal, boqItems, etc.
Task 2.3	Build the /calculator page UI. Must include: ProjectTypeSelector, floor area input, custom items table (add/remove rows), import duties toggle, profit margin slider, Generate button.

Verify	Open /calculator in browser. Fill in form. Click generate. See a table of BOQ items and totals appear on screen.
Task 2.4	Implement Upstash rate limiting on calculator.generate for free tier (3 per day). Show a paywall modal when limit is hit.
Verify	Generate 4 times as a free user. 4th attempt shows paywall modal with Upgrade button.

Phase 3: Projects & PDF (Days 8–12)

Task 3.1	Implement project.create — save the estimate result to the Project and BoqItem tables. Auto-save on every Generate.
Verify	Generate an estimate. Check Supabase — Project row exists with correct totalCostKsh.
Task 3.2	Implement pdf-generator.ts using the layout in Section 7. Connect to project.regeneratePdf procedure.
Verify	Generate a project. Click Download PDF. A PDF opens with the correct BOQ table, totals, and branding.
Task 3.3	Build /projects page with TanStack Table. Show name, type, location, total, date, status, PDF download button.
Verify	Save 3 projects. /projects shows all 3 with correct data.
Task 3.4	Build /projects/[id] page. Show full BOQ, all line items, totals, PDF download, status selector.
Verify	Click a project from the list. Full BOQ renders correctly.

Phase 4: Payments (Days 13–17)

Task 4.1	Implement mpesa.ts from Section 9 in sandbox mode.
Verify	Call initiateSTKPush() directly with a Safaricom test number. Sandbox responds with CheckoutRequestID.
Task 4.2	Implement /api/webhooks/mpesa callback handler from Section 9.3. Use Ngrok to expose locally.
Verify	In Safaricom sandbox simulator, trigger a successful payment. Check Supabase — Subscription.status changes to ACTIVE.
Task 4.3	Implement subscription.upgrade tRPC procedure. Build /billing page with plan cards and Upgrade button.
Verify	Click Upgrade on /billing. Phone receives (simulated) STK push. After callback, /billing shows Pro badge.
Task 4.4	Implement bank transfer fallback. Show bank details (account number, reference) after user selects 'Bank Transfer'.
Verify	Select bank transfer. Bank details display with the correct payment reference format (EQ-{userId}).

Task 4.5	Build /admin/rates page. Implement all admin.* tRPC procedures. Protect with ADMIN role check.
Verify	Log in with ADMIN account. /admin/rates shows all material rates. Edit a rate — change persists immediately.

Phase 5: Polish & Launch (Days 18–28)

Task 5.1	Build landing page /. Must include: headline, product demo screenshot/GIF, feature list, pricing table, CTA buttons.
Verify	Lighthouse score > 80 for Performance, > 90 for SEO.
Task 5.2	Implement Inngest renewal reminder function. 7 days before expiry, send reminder email via Resend.
Verify	Manually trigger the function with a test subscription. Email arrives in inbox.
Task 5.3	Install PostHog. Add events: calculator_opened, estimate_generated, pdf_downloaded, upgrade_clicked, upgrade_success.
Verify	Open the calculator. Check PostHog Live Events — calculator_opened appears within 5 seconds.
Task 5.4	Deploy to Vercel. Configure all environment variables (Section 10). Set up custom domain engiquote.co.ke with SSL.
Verify	https://engiquote.co.ke loads. Sign up works. Estimate generates. PDF downloads.
Task 5.5	Conduct end-to-end test: New user → sign up → generate estimate → download PDF → upgrade via M-Pesa sandbox → verify Pro badge.
Verify	All steps complete without errors. No console errors in production.

12. Revenue Projections & Launch Marketing Plan

12.1 Pricing Rationale

KSh 4,999/month is approximately $38 USD — significantly below global equivalents (PlanSwift: $1,595/year, Buildertrend: $499/month). It is accessible to a Kenyan contractor who bills KSh 2M+ on a single project. Position the price as 'less than a day's labour'.

12.2 Revenue Model

Milestone	Pro Subscribers	MRR (KSh)	Timeline
Soft Launch (friends/network)	10	49,990	Week 6
Month 2 (WhatsApp groups)	30	149,970	Week 10
Month 3 (LinkedIn + referral)	60	299,940	Week 14
Month 6 (organic + 1 enterprise)	100	514,900	Week 26

12.3 Launch Channels (Ordered by ROI)

6. WhatsApp contractor groups — Post a 60-second Loom demo showing a BOQ generated in 3 minutes. Include a free trial link. Target: NCA registered contractor groups in Nairobi, Mombasa, Kisumu.
7. LinkedIn — Post weekly: 'Estimate this project in 5 minutes' challenge with real Kenyan project types. Tag engineering firms and county governments.
8. Facebook Engineering Kenya groups — Offer 50% off first month (KSh 2,499) to first 30 signups. Creates urgency and word of mouth.
9. NCA (National Construction Authority) events — Attend quarterly forums. One in-person demo converts better than 100 cold messages.
10. Quantity Surveyor firms — Target firms with 5–20 staff for Enterprise tier. One enterprise deal = 3× the Pro revenue.
11. SEO — Target: 'BOQ generator Kenya', 'construction cost estimator Kenya', 'building estimate software Nairobi'. These are low competition, high intent keywords.

13. Security & Quality Checklist

This checklist must be completed before going to production. The agent should implement all items marked [MVP]. Items marked [POST-LAUNCH] can be deferred.

Security Control	Priority	Implementation
Zod validation on all tRPC inputs (no raw request data enters DB)	MVP	Already in Section 6 schema
Clerk auth middleware on all (app) routes — redirect to /sign-in if unauthenticated	MVP	Clerk middleware.ts
Row-level security: projects query always filters by userId	MVP	tRPC context + Prisma where clause
Admin routes check role === ADMIN — return 403 otherwise	MVP	tRPC middleware
M-Pesa amount verification in callback (prevent paying KSh 1 for Pro)	MVP	Section 9.3 callback handler
Clerk webhook signature verification (prevent fake user.created events)	MVP	svix library from Clerk docs
Upstash rate limiting on calculator.generate (3/day free, 100/day pro)	MVP	Section 8 — rate-limiter.ts
HTTPS only — enforce via Vercel (automatic)	MVP	Vercel config
Environment variables never exposed to client (no NEXT_PUBLIC_ on secrets)	MVP	Code review
Sentry error tracking with source maps	MVP	Sentry Next.js SDK
OWASP ZAP scan before go-live	POST-LAUNCH	Run ZAP against staging URL
80%+ unit test coverage on calculator engine	POST-LAUNCH	Jest + @testing-library
E2E tests for signup → generate → pay flow	POST-LAUNCH	Playwright
GDPR-style privacy policy and data deletion endpoint	POST-LAUNCH	Legal page + user.delete webhook

14. What This Blueprint Adds Over v1–v3.5

Feature / Specification	Previous Blueprints	This Blueprint (v4)
Exact file creation order	Listed files, no order	28 ordered tasks with verify steps
Database schema	Good — 5 models	6 models + PaymentLog audit trail
Seed data (material rates)	Not included	28 real Kenyan rates with import duties
M-Pesa implementation	Described in 5 bullet points	Full code: token, STK push, callback, amount verify
PDF specification	"Render a table"	Exact column widths, styles, layout, upload flow
Calculator templates	Referenced but not defined	BUILDING and RENOVATION templates with real quantities
Environment variables	12 vars, no descriptions	21 vars with source, example values, notes
Security checklist	5 bullet points	14-item checklist with implementation guide
Rate limiting implementation	Mentioned: use Upstash	Actual implementation in tRPC procedure
Marketing plan	3 bullet points	6 channels ordered by ROI with tactics
Revenue projections	"40-60 Pro subscribers"	Month-by-month table with realistic milestones
Error handling strategy	Not mentioned	Try/catch on M-Pesa, PDF upload, calculator engine

⚑ Final Message to Agent

Read this entire document before writing a single line of code. Follow Section 11 (Build Order) exactly. When in doubt about any implementation detail, refer back to the relevant section — the answers are here. Do not invent solutions for problems this document already solves.

— End of Blueprint — EngiQuoteKE v4.0 — March 2026 —